

I.1)

$$\frac{dx}{dt} = 2x - 4y$$

$$\frac{dy}{dt} = 2x - 2y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} 2 & -4 \\ 2 & -2 \end{bmatrix}$$

EIGENVALUES

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & -4 \\ 2 & -2-\lambda \end{vmatrix} = (2-\lambda)(-2-\lambda) + 8 = \lambda^2 + 4$$

$$\lambda^2 + 4 = 0 \Rightarrow \lambda = \pm 2i$$

$$\lambda_1 = +2i$$

$$\lambda_2 = -2i$$

DELTA-TAU PLANE

$$\Delta = \lambda_1 \lambda_2 = 4$$

(check the Δ - τ diagram)

$\Rightarrow$ EP (0,0) is a CENTRE

$$\tau = 2i - 2i = 0$$

EIGENVECTORS

$$A - \lambda_1 I = \begin{pmatrix} 2-2i & -4 \\ 2 & -2-2i \end{pmatrix} \Rightarrow \underline{V_1} = \begin{pmatrix} 4 \\ 2-2i \end{pmatrix}$$

$$A - \lambda_2 I = \begin{pmatrix} 2+2i & -4 \\ 2 & -2+2i \end{pmatrix} \Rightarrow \underline{V_2} = \begin{pmatrix} 4 \\ 2+2i \end{pmatrix}$$

GENERAL SOLUTION

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{\lambda_1 t} \underline{v}_1 + C_2 e^{\lambda_2 t} \underline{v}_2 = \\ = C_1 e^{2ti} \begin{pmatrix} 4 \\ 2-2i \end{pmatrix} + C_2 e^{-2ti} \begin{pmatrix} 4 \\ 2+2i \end{pmatrix}$$

$$x(t) = 4C_1 (\cos(2t) + i \sin(2t)) + 4C_2 (\cos(2t) - i \sin(2t))$$

$$y(t) = (2-2i)C_1 (\cos(2t) + i \sin(2t)) + (2+2i)C_2 (\cos(2t) + i \sin(2t))$$

For both equations, let us factorize the real part and the imaginary part

$$x(t) = 4(C_1 + C_2) \cos(2t) + 4(C_1 - C_2) \sin(2t) i$$

$$y(t) = 2(C_1 + C_2) (\cos(2t) + \sin(2t)) + 2(C_2 - C_1) (\cos(2t) - \sin(2t)) i$$

$$a_1 = 2(C_1 + C_2)$$

$$a_2 = 2(C_1 - C_2) i$$

$$x(t) = 2a_1 \cos(2t) + 2a_2 \sin(2t)$$

$$y(t) = a_1 (\cos(2t) + \sin(2t)) - a_2 (\cos(2t) - \sin(2t))$$

I.2)

$$\frac{dx}{dt} = 3x + y$$

$$\frac{dy}{dt} = -x + 5y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 5 \end{bmatrix}$$

EIGENVALUES

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ -1 & 5-\lambda \end{vmatrix} = (3-\lambda)(5-\lambda) + 1 = \lambda^2 - 8\lambda + 16$$

$$(\lambda - 4)^2 = 0 \Rightarrow \lambda = 4$$

DELTA-TAU PLANE

$$\Delta = 16$$

(check the Δ - τ diagram)

$$\tau = 8$$

$\Rightarrow$ EP (0,0) IS AN UNSTABLE DEGENERATE NODE

$$\tau^2 - 4\Delta = 0$$

EIGENVECTOR

$$A - \lambda I = \begin{pmatrix} 3-4 & 1 \\ -1 & 5-4 \end{pmatrix} = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \Rightarrow \underline{v} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

look for the unknown vector p

$$(A - \lambda I)\underline{p} = \underline{v} \Rightarrow \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} p_x \\ p_y \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$$

$$-p_x + p_y = -1 \quad \underline{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

GENERAL SOLUTION

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{2t} \underline{v} + C_2 e^{2t} (\underline{v}t + \underline{p}) =$$

$$= C_1 e^{4t} \begin{pmatrix} -1 \\ -1 \end{pmatrix} + C_2 e^{4t} \left(\begin{pmatrix} -1 \\ -1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right) =$$

$$\boxed{\begin{aligned} x(t) &= [C_2(1-t) - C_1] e^{4t} \\ y(t) &= -(C_1 + C_2 t) e^{4t} \end{aligned}}$$

PS2

I.3)

$$\frac{dx}{dt} = 4x + 2y$$

$$\frac{dy}{dt} = 3x - y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} 4 & 2 \\ 3 & -1 \end{bmatrix}$$

EIGEN VALUES

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & 2 \\ 3 & -1-\lambda \end{vmatrix} = (4-\lambda)(-1-\lambda) - 6 = -4 - 4\lambda + \lambda + \lambda^2 - 6 = \lambda^2 - 3\lambda - 10$$

$$\lambda^2 - 3\lambda - 10 > 0 \Rightarrow \lambda_1 = -2 \text{ and } \lambda_2 = 5$$

DELTA-TAU PLANE

$$\Delta = \lambda_1 \lambda_2 = -10$$

(check the Δ - τ diagram)

$$\tau = \lambda_1 + \lambda_2 = 3$$

 $\Rightarrow$ E.P. (0,0) is a SADDLE POINTEIGENVECTORS

$$A - \lambda_1 I = \begin{pmatrix} 4+2 & 2 \\ 3 & -1+2 \end{pmatrix} = \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix} \Rightarrow \underline{v}_1 = \begin{pmatrix} -2 \\ 6 \end{pmatrix}$$

$$A - \lambda_2 I = \begin{pmatrix} 4-5 & 2 \\ 3 & -1-5 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 3 & -6 \end{pmatrix} \Rightarrow \underline{v}_2 = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

(trick)

GENERAL SOLUTION

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = C_1 e^{\lambda_1 t} \underline{v}_1 + C_2 e^{\lambda_2 t} \underline{v}_2 = C_1 e^{-2t} \begin{pmatrix} -2 \\ 6 \end{pmatrix} + C_2 e^{5t} \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

$$x(t) = -2C_1 e^{-2t} - 2C_2 e^{5t}$$

$$y(t) = 6C_1 e^{-2t} - C_2 e^{5t}$$

II. 1)

$$\frac{dx}{dt} = 4x - 2y$$

$$\frac{dy}{dt} = 3x - y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} 4 & -2 \\ 3 & -1 \end{bmatrix}$$

EIGENVALUES

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 4-\lambda & -2 \\ 3 & -1-\lambda \end{vmatrix} = (4-\lambda)(-1-\lambda) + 6 = -4 - 4\lambda + \lambda + \lambda^2 + 6 = \lambda^2 - 3\lambda + 2$$

$$\lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda_1 = 1 \text{ and } \lambda_2 = 2$$

DELTA - TAU PLANE

$$\Delta = \lambda_1 \lambda_2 = 2$$

 $\Rightarrow$ (check the Δ - τ diagram) $\tau^2 - 4\Delta = 1$

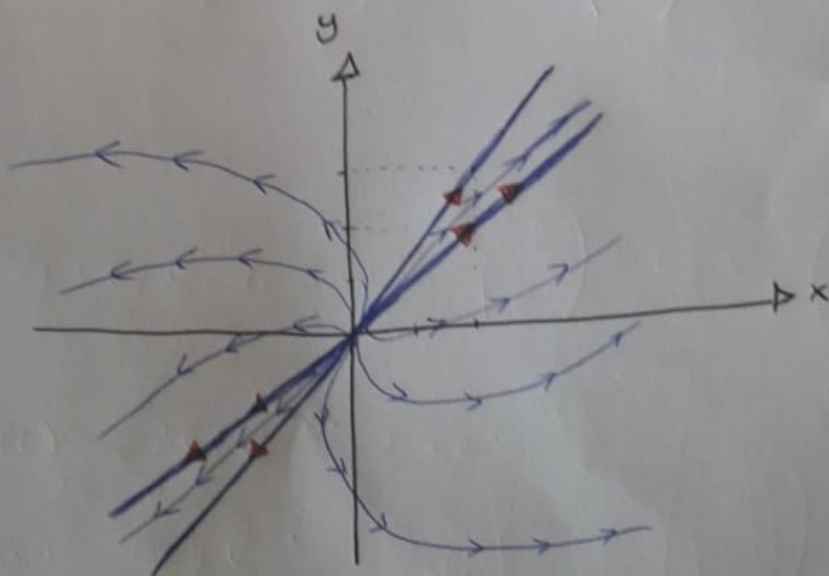
$$\tau = \lambda_1 + \lambda_2 = 3$$

E.P. (0,0) IS AN UNSTABLE NODEEIGENVECTORS

$$A - \lambda_1 I = \begin{pmatrix} 4-1 & -2 \\ 3 & -1-1 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ 3 & -2 \end{pmatrix} \Rightarrow \underline{v}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

(twice)

$$A - \lambda_2 I = \begin{pmatrix} 4-2 & -2 \\ 3 & -1-2 \end{pmatrix} = \begin{pmatrix} 2 & -2 \\ 3 & -3 \end{pmatrix} \Rightarrow \underline{v}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

PHASE PORTRAIT

PS2

II.2) $\frac{dx}{dt} = x - 3y$
 $\frac{dy}{dt} = 4x - 2y$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} 1 & -3 \\ 4 & -2 \end{bmatrix}$$

EIGENVALUES

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & -3 \\ 4 & -2-\lambda \end{vmatrix} = (1-\lambda)(-2-\lambda) + 12 = -2 - \lambda + 2\lambda + \lambda^2 + 12 = \lambda^2 + \lambda + 10$$

$$\lambda^2 + \lambda + 10 = 0 \Rightarrow \lambda_{1,2} = \frac{-1 \pm \sqrt{1-40}}{2} = -\frac{1}{2} \pm \frac{\sqrt{39}}{2} i$$

DELTA-TAU PLANE

$$\Delta = 10$$

$$\tau = -1$$

$$\tau^2 - 4\Delta = -39$$

(check the diagram)

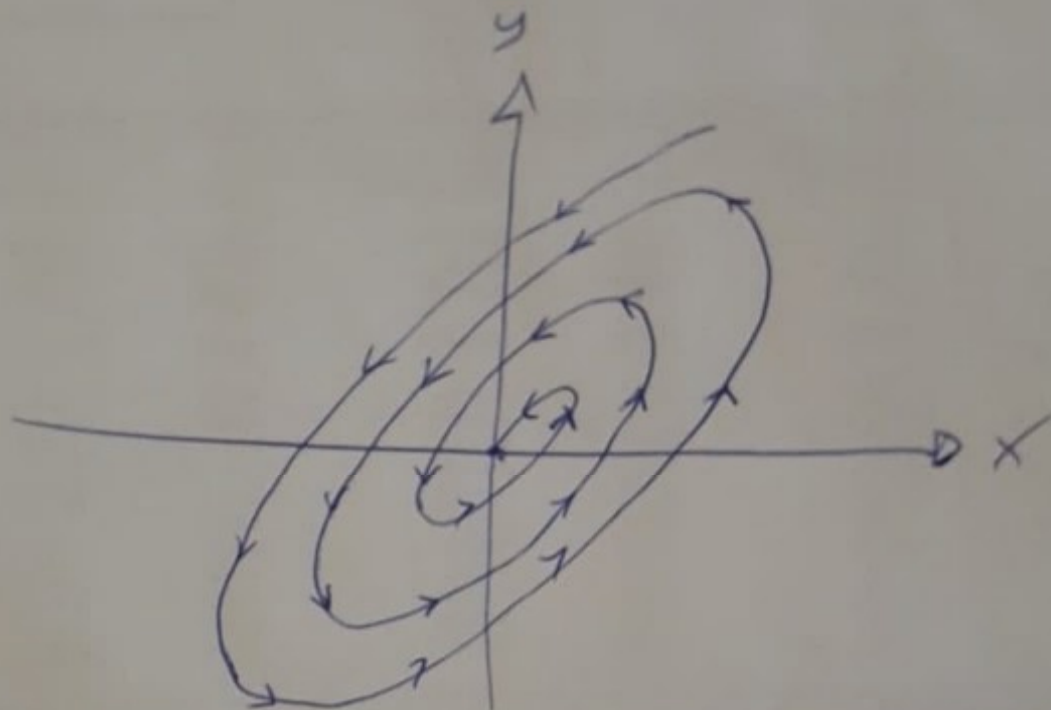
E.P. (0,0) is a STABLE SPIRAL

DIRECTION OF ROTATION

Let $x = 0 + \epsilon$
 $y = 0$ for $\epsilon > 0 \Rightarrow \dot{x} = \epsilon > 0$
 $\dot{y} = 4\epsilon > 0 \Rightarrow$



PHASE PORTRAIT



II.3)

$$\frac{dx}{dt} = 3x - y$$

$$\frac{dy}{dt} = x + y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$$

EIGENVALUES

$$\det(A - \lambda I) = 0$$

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & -1 \\ 1 & 1-\lambda \end{vmatrix} = (3-\lambda)(1-\lambda) + 1 = 3 - 3\lambda - \lambda + \lambda^2 + 1 = \lambda^2 - 4\lambda + 4$$

$$(\lambda^2 - 4\lambda + 4) = 0 \Rightarrow \lambda = 2$$

DELTA-TAU PLANE

$$\Delta = 4$$

$$\tau = 4$$

$$\tau^2 - 4\Delta = 0$$

 $\Rightarrow$

(check the diagram)

E.P. (0,0) IS AN UNSTABLE DEGENERATE NODE

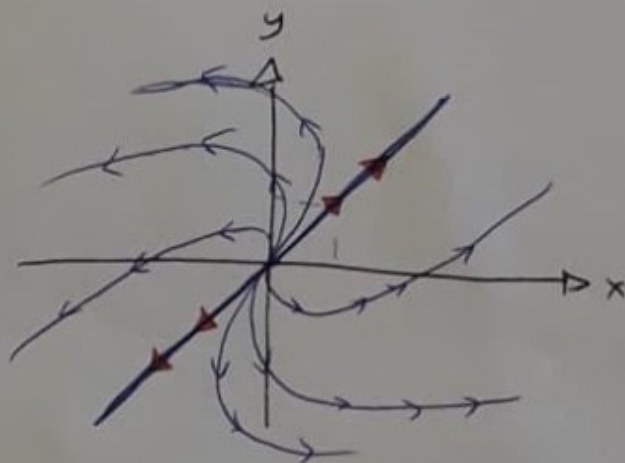
EIGEN VECTOR

$$A - \lambda I = \begin{pmatrix} 3-2 & -1 \\ 1 & 1-2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \Rightarrow \underline{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (\text{triple})$$

DIRECTION OF DYNAMICS

$$\text{Let } \begin{matrix} x=0 \\ y=\varepsilon \end{matrix} \text{ for } \varepsilon > 0 \Rightarrow \begin{matrix} \dot{x} = -\varepsilon < 0 \\ \dot{y} = \varepsilon > 0 \end{matrix} \Rightarrow \begin{matrix} \nearrow \\ \nwarrow \end{matrix} \begin{matrix} (0, \varepsilon) \end{matrix}$$

$$\text{Let } \begin{matrix} x=\varepsilon \\ y=-\varepsilon \end{matrix} \text{ for } \varepsilon > 0 \Rightarrow \begin{matrix} \dot{x} = \varepsilon > 0 \\ \dot{y} = -\varepsilon < 0 \end{matrix} \Rightarrow \begin{matrix} \searrow \\ \swarrow \end{matrix} \begin{matrix} (\varepsilon, -\varepsilon) \end{matrix}$$

PHASE PORTRAIT

III.1)

$$\frac{dx}{dt} = 3x + ay$$

$$\frac{dy}{dt} = x + y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} 3 & a \\ 1 & 1 \end{bmatrix}$$

DELTA-TAU PLANE

$$\tau = \text{tr}(A) = 3 + 1 = 4$$

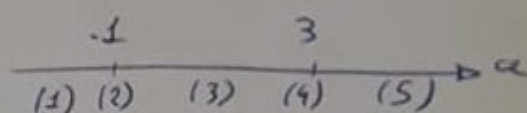
$$\Delta = \det(A) = 3 - a$$

$$\tau^2 - 4\Delta = 16 - 4(3 - a) = 4(1 + a)$$

CRITICAL POINTS

$$\Delta = 0 \Leftrightarrow a = 3$$

$$\tau^2 - 4\Delta = 0 \Leftrightarrow 4(1 + a) = 0 \Leftrightarrow a = -1$$

NATURE OF E.P. (0,0)

- (1) $a < -1$: unstable spiral(focus)
- (2) $a = -1$: degenerate unstable node
- (3) $-1 < a < 3$: unstable node
- (4) $a = 3$: whole line of unstable fixed point (unstable equilibrium set)
- (5) $a > 3$: saddle point

PS3

$$\text{III.2) } \frac{dx}{dt} = ax - y$$

$$\frac{dy}{dt} = -4x - 2y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} a & -1 \\ 2 & -4 \end{bmatrix}$$

DELTA-TAU PLANE

$$\tau = \text{tr}(A) = a - 4$$

$$\Delta = \det(A) = -4a + 2 = 2(1 - 2a)$$

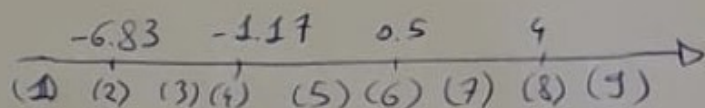
$$\tau^2 - 4\Delta = (a-4)^2 - 8(1-2a) = a^2 - 8a + 16 - 8 + 16a = a^2 + 8a + 8$$

CRITICAL POINTS

$$\tau = 0 \Leftrightarrow a = 4$$

$$\Delta = 0 \Leftrightarrow a = \frac{1}{2} = 0.5$$

$$\tau^2 - 4\Delta = 0 \Leftrightarrow a^2 + 8a + 8 = 0 \Rightarrow a = \frac{-8 \pm \sqrt{64 - 32}}{2} = \begin{cases} -4 + 2\sqrt{2} \approx -1.17 \\ -4 - 2\sqrt{2} \approx -6.83 \end{cases}$$

NATURE OF E.P. (0,0)(1) $a < -4 - 2\sqrt{2}$: stable node(2) $a = -4 - 2\sqrt{2}$: degenerate stable node(3) $-4 - 2\sqrt{2} < a < -4 + 2\sqrt{2}$: stable spiral(4) $a = -4 + 2\sqrt{2}$: degenerate stable node(5) $-4 + 2\sqrt{2} < a < \frac{1}{2}$: stable node(6) $a = \frac{1}{2}$: non isolated fixed points (stable equilibrium set)(7) $\frac{1}{2} < a < 4$:(8) $a = 4$:(9) $a > 4$:

saddle point
saddle point
saddle point

 $a > \frac{1}{2}$: saddle point

PS3

III.3)

$$\frac{dx}{dt} = -4x - 2y$$

$$\frac{dy}{dt} = ax + y$$

COEFFICIENT MATRIX

$$A = \begin{bmatrix} -4 & -2 \\ a & 1 \end{bmatrix}$$

DELTA - TAU PLANE

$$\tau = \text{tr}(A) = -4 + 1 = -3$$

$$\Delta = \det(A) = -4 + 2a = 2(a-2)$$

$$\tau^2 - 4\Delta = 9 - 8(a-2) = -8a + 25$$

CRITICAL POINTS

$$\Delta = 0 \Leftrightarrow a = 2$$

$$\tau^2 - 4\Delta = 0 \Leftrightarrow a = \frac{25}{8} = 3.125$$

$$\begin{array}{ccccccc} & & 2 & & 3.125 & & \\ (1) & (2) & (3) & (4) & (5) & & \end{array} \rightarrow$$

NATURE OF E.P. (0,0)(1) $a < 2$: saddle point(2) $a = 2$: whole line of stable e.p. (stable equilibrium)(3) $2 < a < \frac{25}{8}$: stable node(4) $a = \frac{25}{8}$: degenerate stable node(5) $a > \frac{25}{8}$: stable spiral